Prodigy Anesthesia Quick Review

Cardioversion

What is cardioversion?	Cardioversion is the application of electrical energy to a patient to convert a cardiac rhythm such as atrial fibrillation, atrial flutter, or stable ventricular tachycardia to a normal rhythm. The electrical discharge is timed to coincide with the R wave of the QRS complex. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1269-1270.*
What is the difference between cardioversion and defibrillation?	Cardioversion is typically a scheduled and planned procedure (although it can be performed emergently as well) that uses synchronized electrical energy to convert a patient's cardiac rhythm. Defibrillation is an emergent procedure that is not synchronized to the patient's cardiac rhythm and requires a much larger amount of electrical energy. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1270.*
How does the preoperative preparation for a cardioversion differ from that of a patient undergoing general anesthesia for surgery?	It doesn't. The patient should undergo as thorough a preoperative evaluation as any other patient. NPO status should be as rigorously verified as for any other surgery because the risk of aspiration is just as likely. If an emergent cardioversion is necessary and the patient is to be anesthetized for the procedure, then tracheal intubation is required. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1270.*
What energy level is used to 'shock' the patient back into a normal rhythm?	Ideally, 50-100 Joules are used to cardiovert the patient. In the event the rhythm does not convert, the energy level is increased to a maximum level of 360 Joules. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1270.*
Is muscle relaxation required for cardioversion?	No, but if the patient is not NPO, a rapid-sequence induction using a muscle relaxant prior to intubation may be required. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1270.*
How is an anesthetic for cardioversion generally administered?	After standard anesthesia monitors are applied, the patient is pre-oxygenated via an ambu bag or via mask using an anesthesia machine and circuit. The patient is administered an intravenous induction agent. Once the eyelash reflex is abolished, positive-pressure ventilation is halted. An 'all clear' signal is announced by the person performing the cardioversion and the synchronized shock is delivered. The patient is assisted in ventilation until normal swallowing and coughing reflexes return and the care is turned over to the recovery nurse for postanesthesia and cardiac monitoring. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1270.*

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